LAKE WATER

Cool upland lake water contains very few dissolved minerals that could feed plants. This keeps the lakes clear and almost weed-free (as above). Lowland lake water is usually warmer and rich in nutrients. It supports large numbers of microscopic plankton, which make the water appear cloudy.

LAKE EVOLUTION

Lowland lakes with nutrient-rich water support a wide range of plant life, including plants such as reeds that grow in the muddy lake fringes. The reeds tend to trap silt, which builds up to allow more plants to take root. Over time, this process turns the lake into waterlogged marshland. Trees take root and the marshland becomes wet woodland. Eventually the lake may disappear altogether.

▲ DISAPPEARING LAKE
Reeds and trees are gradually taking over this small lake.

SALT AND SODA LAKES

Lake water contains salty minerals that have been carried into it by streams. Normally, the minerals are carried out of the lake by streams that flow to the sea. However, in hot climates, the water evaporates as fast as the lake fills. As the water evaporates, it leaves the minerals behind, where they form crystals on rocks (below) and make the water very salty. The result is a salt or soda lake, depending on the minerals.

▼ SALT LAKE
The water in the Dead Sea in the Jordan Rift Valley is so salty that no plants or animals can survive in it.